

NOVARA CAMERI, Italy

WMO: 160640

Lat: 45.530N

Lon: 8.667E

Elev: 179

StdP: 99.20

Time Zone: 1.00 (EUC)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.2	(c) -5.8	(d) -18.0	(e) 0.8	(f) 9.8	(g) -13.8	(h) 1.2	(i) 7.9	(j) 9.9	(k) 6.6	(l) 8.1	(m) 7.5	(n) 0.8	(o) 10	(p) 0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 31.1	(d) 21.7	(e) 30.1	(f) 21.5	(g) 29.0	(h) 20.7	(i) 24.0	(j) 29.1	(k) 23.1	(l) 28.0	(m) 22.2	(n) 26.6	(o) 1.9	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.3	26.4	21.8	16.8	25.8	20.9	15.9	24.7	73.1	29.0	69.4	27.8	65.9	27.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	6.1	4.9	-10.4	32.8	1.4	0.9	-11.3	33.4	-12.1	34.0	-12.9	34.5	-13.9	35.2
			-10.4	25.0	1.3	1.4	-11.4	26.0	-12.1	26.9	-12.8	27.7	-13.8	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	2.1	4.3	8.8	11.5	17.1	20.2	22.6	22.7	18.2	12.5	6.5	2.4	(6)	
	DBStd	7.87	2.55	3.17	2.92	2.80	2.77	2.86	2.30	2.28	2.49	3.20	3.15	2.97	(7)	
	HDD10.0	839	244	161	57	17	1	0	0	0	0	14	110	236	(8)	
	HDD18.3	2532	503	394	296	206	56	14	1	2	32	181	354	494	(9)	
	CDD10.0	1735	0	1	19	61	219	305	391	394	247	92	6	0	(10)	
	CDD18.3	386	0	0	0	0	17	69	133	137	29	1	0	0	(11)	
	CDH23.3	3298	0	0	2	4	157	603	1208	1133	186	4	0	0	(12)	
	CDH26.7	928	0	0	0	0	19	142	384	365	18	1	0	0	(13)	
(14) Wind	WSAvg	1.6	1.3	1.6	2.1	2.2	2.1	1.9	1.6	1.5	1.4	1.4	1.4	1.2	(14)	
(15) Precipitation	PrecAvg	1087	72	82	99	82	122	94	65	98	89	119	122	55	(15)	
	PrecMax	1689	206	268	279	227	332	194	160	240	265	337	225	129	(16)	
	PrecMin	717	2	0	0	1	22	29	14	7	1	0	8	5	(17)	
	PrecStd	297	52	67	74	56	84	51	39	58	73	102	68	37	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	17.9	22.2	23.9	28.8	32.1	32.2	32.2	28.8	23.9	16.9	12.8	(19)	
		MCWB	5.6	9.2	12.0	13.9	18.1	21.4	22.2	21.8	19.2	17.5	10.0	5.5	(20)	
	2%	DB	10.0	14.9	19.8	21.8	26.8	29.9	31.1	31.1	27.0	21.2	14.8	10.0	(21)	
		MCWB	5.3	8.0	11.3	12.3	17.3	20.7	21.7	21.9	18.1	15.3	10.0	5.5	(22)	
	5%	DB	8.2	12.9	17.9	20.0	25.0	28.2	30.1	30.1	25.8	19.5	13.1	8.2	(23)	
		MCWB	4.8	6.8	10.4	12.0	16.6	20.0	21.6	21.9	17.8	14.4	10.3	6.2	(24)	
	10%	DB	6.9	10.9	15.9	18.2	23.6	26.9	29.0	29.0	24.0	17.9	11.9	7.3	(25)	
		MCWB	4.6	6.2	9.5	10.8	16.3	19.2	20.8	21.2	17.3	14.3	9.8	5.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	10.8	14.0	15.5	20.0	23.8	25.8	24.8	21.1	18.3	13.2	9.2	(27)	
		MCDB	9.6	15.0	19.7	20.5	25.8	28.3	30.2	29.7	25.3	21.4	14.7	9.7	(28)	
	2%	WB	6.7	9.1	12.7	14.1	18.7	22.6	24.4	23.8	20.1	17.1	12.1	8.0	(29)	
		MCDB	7.9	13.4	17.4	18.6	24.3	27.7	29.4	29.2	23.9	19.6	13.3	8.4	(30)	
	5%	WB	5.7	7.8	11.6	13.1	17.7	21.4	23.5	22.9	19.4	16.2	11.2	7.1	(31)	
		MCDB	6.8	11.0	16.1	17.4	22.6	26.1	28.1	27.8	23.3	18.2	12.3	7.6	(32)	
	10%	WB	4.7	6.7	10.4	12.3	16.9	20.4	22.4	22.1	18.6	15.3	10.3	5.9	(33)	
		MCDB	5.8	9.2	14.1	16.6	22.0	25.2	26.5	26.5	22.2	17.1	11.3	6.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	10.6	11.8	10.7	10.4	10.8	11.1	10.9	10.4	8.8	8.4	8.1	(35)	
		MCDBR	11.7	15.1	16.0	14.3	13.1	12.2	12.3	12.3	12.6	11.5	11.0	10.2	(36)	
	5% WB	MCWBR	7.9	9.1	8.7	6.8	5.8	5.5	5.6	5.5	5.4	6.3	7.0	6.9	(37)	
		MCWBR	8.6	12.5	14.0	11.2	11.6	11.4	11.3	11.6	9.8	9.3	6.9	6.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.343	0.372	0.422	0.468	0.469	0.489	0.470	0.464	0.452	0.434	0.376	0.336	(40)	
		taud	2.355	2.259	2.138	2.049	2.104	2.091	2.146	2.166	2.157	2.162	2.288	2.381	(41)	
	Ebn at Noon		747	790	796	792	806	789	800	788	757	704	690	712	(42)	
		Edh at Noon	78	102	133	158	154	157	147	139	129	112	83	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.24	3.58	4.50	5.54	6.39	6.51	5.51	4.13	2.43	1.44	1.21	(44)	
	RadStd		0.21	0.33	0.46	0.51	0.45	0.38	0.41	0.34	0.28	0.38	0.25	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (76 neighbors)	+0.32	N/A	N/A	+0.68	+0.40	+0.31	N/A	N/A	+57	+10	(47)

Nomenclature: See separate page